

Topic 6 – Plant structures and their functions

Students should:	Maths skills
6.1 Describe photosynthetic organisms as the main producers of food and therefore biomass	
6.2 Describe photosynthesis in plants and algae as an endothermic reaction that uses light energy to react carbon dioxide and water to produce glucose and oxygen	
6.3 Explain the effect of temperature, light intensity and carbon dioxide concentration as limiting factors on the rate of photosynthesis	2c, 2d, 2g 4a, 4c
6.4 Explain the interactions of temperature, light intensity and carbon dioxide concentration in limiting the rate of photosynthesis	4b, 4c, 4d
6.5 <i>Core Practical: Investigate the effect of light intensity on the rate of photosynthesis</i>	2c, 2f, 2g 4a, 4c
6.6 Explain how the rate of photosynthesis is directly proportional to light intensity and inversely proportional to the distance from a light source, including the use of the inverse square law calculation	2g 3a 4a, 4b, 4c, 4d
6.7 Explain how the structure of the root hair cells is adapted to absorb water and mineral ions	
6.8 Explain how the structures of the xylem and phloem are adapted to their function in the plant, including: a lignified dead cells in xylem transporting water and minerals through the plant b living cells in phloem using energy to transport sucrose around the plant	
6.9 Explain how water and mineral ions are transported through the plant by transpiration, including the structure and function of the stomata	
6.10 Describe how sucrose is transported around the plant by translocation	
6.11B Explain how the structure of a leaf is adapted for photosynthesis and gas exchange	2d 5c
6.12 Explain the effect of environmental factors on the rate of water uptake by a plant, to include light intensity, air movement and temperature	1a, 1c 2b, 2c 4a, 4b, 4c, 4d
6.13 Demonstrate an understanding of rate calculations for transpiration	1a, 1c 2b, 2c 4a, 4b, 4c, 4d

Students should:	Maths skills
6.14B Explain how plants are adapted to survive in extreme environments including the effect of leaf size and shape, the cuticle and stomata	2d 5c
6.15B Explain how plant hormones control and coordinate plant growth and development, including the role of auxins in phototropisms and gravitropisms	5a
6.16B Describe the commercial uses of auxins, gibberellins and ethene in plants, including: a auxins in weedkillers and rooting powders b gibberellins in germination, fruit and flower formation and the production of seedless fruit c ethene in fruit ripening	4a, 4c

Use of mathematics

- Carry out rate calculations for chemical reactions (1a and 1c).
- Use simple compound measures such as rate (1a, 1c)
- Plot, draw and interpret appropriate graphs (4a, 4b, 4c and 4d).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Understand the principles of sampling as applied to scientific data (2d).
- Use a scatter diagram to identify a correlation between two variables (2g).
- Understand and use simple compound measures such as the rate of a reaction (1a and 1c).
- Understand and use inverse proportion – the inverse square law and light intensity in the context of factors affecting photosynthesis.
- Use percentiles and calculate the percentage gain and loss of mass (1c).
- Use fractions and percentages (1c).
- Calculate arithmetic means (2b).
- Calculate cross-sectional areas of bacterial cultures and clear agar jelly using πr^2 (5c).
- Carry out rate calculations (1a and 1c).

Suggested practicals

- Investigate the effect of pollutants on plant germination and plant growth.
- Investigate tropic responses.
- Investigate the effect of CO₂ concentration or temperature on the rate of photosynthesis.
- Investigate how the structure of the leaf is adapted for photosynthesis.
- Investigate how the loss of water vapour from leaves drives transpiration.
- Investigate the importance of photoperiodicity in plants.